Supplementary notes

Proof that $Q_x - P_y = 1/\ell$ in the example of the planimeter

We have

$$\hat{N} = \frac{1}{\ell}\langle -(y-b), (x-a)\rangle = \langle P, Q\rangle,$$

where $a = a(x,y)$, $b = b(x,y)$ satisfy

$$\begin{cases} (x-a)^2 + (y-b)^2 = \ell^2 & (1) \\ a^2 + b^2 = \ell^2 & (2) \end{cases}$$

Therefore

$$Q_x - P_y = \frac{1}{\ell}\left((1-a_x) - (-1+b_y)\right)$$

$$= \frac{1}{\ell}(2 - a_x - b_y).$$

To prove that

$$Q_x - P_y = 1/\ell,$$

we have to check that

$$a_x + b_y = 1.$$

We do this using (1) and (2).

Combining (1) and (2), we get:

$$x^2 - 2ax + \cancel{a^2} + y^2 - 2by + \cancel{b^2} = \cancel{\ell^2}$$

$$\Rightarrow x^2 + y^2 = 2ax + 2by \qquad (3)$$

Taking partials from (2):

$$\begin{cases} 2a a_x + 2b b_x = 0 \\ 2a a_y + 2b b_y = 0 \end{cases}$$

$$\Rightarrow \begin{cases} b_x = -(a/b)\, a_x & (2') \\ a_y = -(b/a)\, b_y & (2'') \end{cases}$$

Taking partials from (3):

$$\begin{cases} 2x = 2a_x \cdot x + 2a \cdot 1 + 2b_x y & (3') \\ 2y = 2a_y x + 2b_y \cdot y + 2b \cdot 1 & (3'') \end{cases}$$

Plugging (2') into $b/2$ times (3'):

$$bx = a_x bx + ab - a_x a y$$

$$\Rightarrow a_x = \frac{bx - ab}{bx - ay} \qquad (*)$$

Plugging (2'') into $a/2$ times (3''):

$$ay = -b b_y x + b_y a y + ab$$

$$\Rightarrow b_y = \frac{ay - ab}{ay - bx}. \qquad (**)$$

$\longrightarrow$

Now add ⊛ and ⊛⊛:

$$ax+by = \frac{bx-ab}{bx-ay} + \frac{ay-ab}{ay-bx}$$

$$= \frac{abxy - b^2x^2 - \cancel{a^2by} + \cancel{ab^2x} + abxy - \cancel{ab^2x} - a^2y^2 + \cancel{a^2by}}{abxy - b^2x^2 - a^2y^2 + abxy}$$

$$= 1 \;//$$

↑ This completes the proof!